

Temma Driver

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Introduction

This help file describes how to configure and operate the Temma V2 ASCOM Telescope driver.

The Temma V4 telescope driver supports the ASCOM Telescope Driver Interface standard V4.

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What's new

Temma V4 ASCOM driver features:

- NET Local Server (exe) version
- 100% compliant with ASCOM Platform 7
- Adheres to the ASCOM Telescope V4 Standard.
- Provides custom RA/Dec tracking rates.
- Provides PulseGuide support.

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Revision History

1.0.0 - Initial release ([see Github](#))

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Getting Started

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System requirements

Software

- 1.A computer running in Windows 10/11 (32-bit or 64-bit).
- 2.ASCOM Platform 6 (or later) must be installed.

3..Target Net Framework 4.7.2.

Hardware

1. Takahashi Temma 2, or Temma 2M mount.

2. A computer with a serial port or USB/Ethernet to serial port converter.

Note: Users should avoid Keyspan USA-19HS serial adapters, BSOD crashes occur when used with the ASCOM Temma driver. The crash dump file points to the Keyspan driver, this issue was reported to Tripp Lite on 12/12/10. BSOD occur from flaws in kernel level drivers, the ASCOM TEMMA driver is user level, indicating a hardware fault or a buggy kernel mode Keyspan driver.

3. A Serial cable to connect a PC to the telescope.

4. Power Source

Select the appropriate power setting from the 'Power' drop-down box. You must have the appropriate 24 volt power supply to run at the 24 volt high slew rates.

High Speed Drive	Working Voltage	
	DC12V	DC24V
EM-400 Temma2	250X	500X
EM-10 Temma2 Jr.	120X	250X
EM-200 Temma2 Jr.	120X	250X
EM-200 Temma2	350X	700X
NJP Temma2	175X	350X
EM-500 Temma2	260X	520X
EM-200M	700X	Not Supported
EM-400M	500X	Not Supported
EM-12M	150X	Not Supported

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Installing

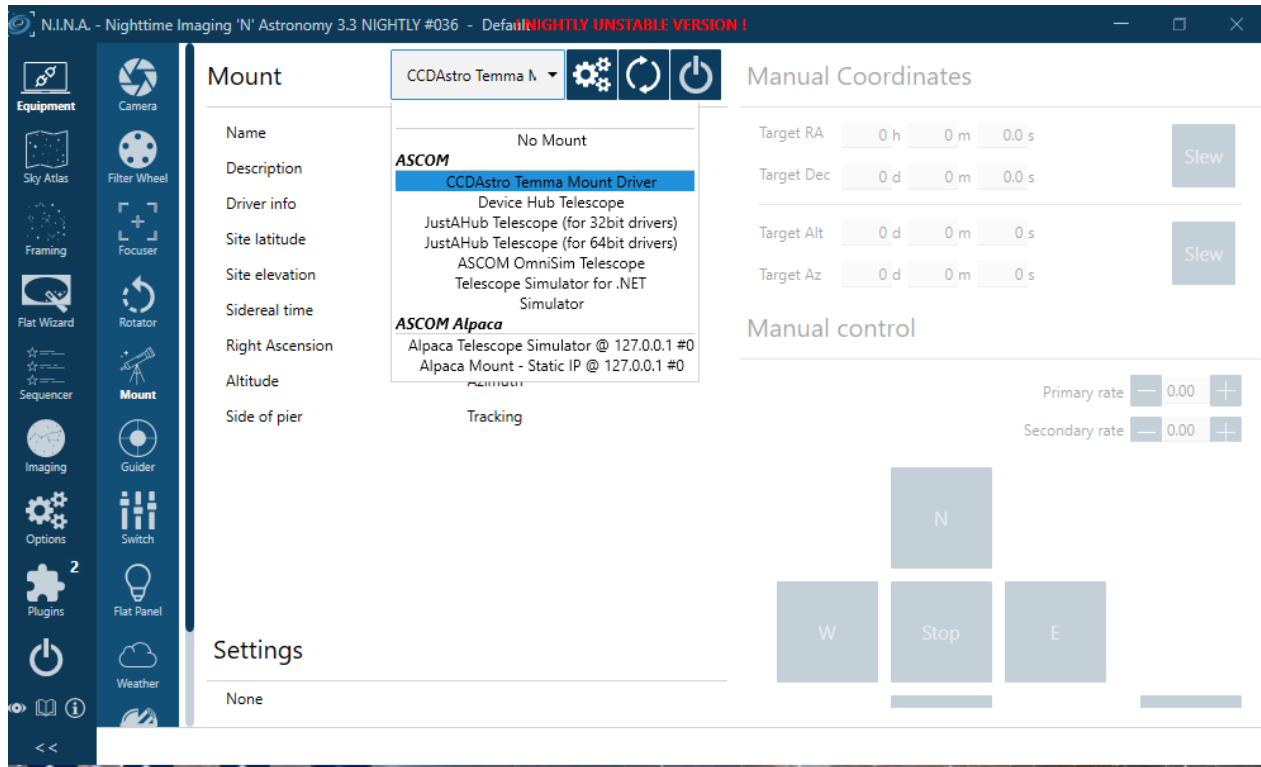
Before installing the Temma V4 telescope driver please make sure you have installed the ASCOM 6.0 (or later) platform. You can download it from <http://www.ascom-standards.org>. You will also need to install the Microsoft .Net Framework 4.7.2 and possibly service packs. More details on the requirements can be found on the ASCOM-Standards site.

You can always install any version of the V4 driver without uninstalling the previously installed version of the V4 driver.

Using the Temma Driver with Client Apps

To use the driver select it in an ASCOM application.

For example, in NINA click the Telescope then choose 'CCDAstro Temma Mount Driver'



Using the Temma Driver with Thesky X

Install the ASCOM X2 Plugin

To use TheSkyX with the ASCOM driver you must first instal the ASCOM Mount/Telescope X2 plugin:

1. Bring up a browser and navigate to <http://www.ascom-standards.org/Downloads/Plugins.htm>

2. Download and install the latest version of Mount/Telescope X2 to ASCOM Converter for TheSky X

Activate TheSky X scripting support:

1. Locate TheSkyX executable. This is (normally in Program Files or Program Files (x86), Software Bisque, TheSky X Professional).
2. Right click and select "Run as Administrator"
3. Answer Yes to the "are you sure" security popup
4. Wait until TheSky X starts up completely
5. Click Telescope Setup in the menu:
6. Click Mount Setup then Choose...
7. Select ASCOM/Telescope Driver for the mount type, then click OK.
8. Now, back in the Imaging System Setup click Mount Setup, Settings... Answer No to the "Do you want to read...?" popup.
9. You will see the usual ASCOM Chooser
10. Select Mount 'CCDAstro Temma Mount Driver'.
11. Click Properties in the Chooser to configure the mount.
12. Close the Chooser.
13. From there it's normal TheSky X operation

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Getting help and reporting driver issues

In order of preference potential driver problems can be reported via several methods:

1. Posting a message to the Yahoo Groups forum for Temma mounts:
<http://tech.groups.yahoo.com/group/ASCOMTemmaDriver/>
2. E-mail to ccdastroinc@gmail.com.

IMPORTANT: When reporting a potential ASCOM driver bug here are some general guidelines:

1. The first aim of a bug report is to let us see the failure with our own eyes. Please give us detailed instructions so that we can make it fail for ourselves.
2. The second aim of a bug report is to describe what went wrong. Describe everything in detail. State what you saw, and also state what you expected to see. Write down the error messages, especially if they have numbers in them. Better yet, take a screenshot!
3. By all means try to diagnose the fault yourself if you think you can, but if you do, you should still

report the symptoms as well.

4. Be ready to provide extra information if we need it, like trace logs.
5. If other programs are involved state the EXACT version number of each program. Usually you can find this information in the programs Help->About menu item.
6. Write clearly. Say what you mean, and make sure it can't be misinterpreted!

Thank you in advance for following these guidelines! For more details please see this link:

[Github](#)

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Temma Mount Setup Window

ASCOM Setup

Telescope Setup window:

Detailed explanations of each field follows in the next sections.

Temma Setup

Mount Setup

Com Port: COM1

Tracking Rate: Sidereal

Voltage: 24V

Model: EM200

☒ Enable Trace Logging

Site Information Setup

Latitude: N 27 0 0

Longitude: W 80 0 0

EI (m): 15

Telescope Orientation

☐ OTA East - Scope Pointing West

☐ OTA West - Scope Pointing East

☒ Counterweight Down ☒ Keep Last Sync

☐ Counterweight West ☐ Ask At Start

Park/Rate Settings

Park Settings

0 Alt - Az 0

☒ Current Position ☐ Slew To

☐ Unpark On Reconnect

Guide Settings

0.50 Ra - Dec 0.50

☐ Send Rate

Telescope Setup

Telescope Aperture (mm): 0

Central Obstruction (mm): 0

Focal Length (mm): 0

☐ Hi-Precision GOTO

☐ Tracking Off on Connection

☐ Warm Before Meridian Flip

OK Cancel

Advanced Features

Automated Mount Safety

The mount safety features is not a failsafe way to protect your equipment from damage, and it is not intended to be a substitute for good judgment or an attendant at the scope. Please use with caution. Keep the checkbox checked to disable these features.

Setup the Parameters:

note: scope movements started via the Temma's hand paddle can not be controlled via the driver's mount safety features. Rules imposed by the driver will only affect slews initiated by the driver. These features are mainly designed to help avoid crashes from the telescope tracking in to the mount, tripod, or other object.

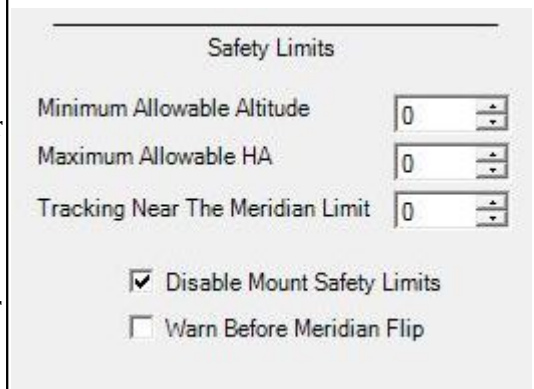
Minimum Allowable Altitude - Input a number representing degrees in altitude that the mount will stop all movement when it is reached or exceeded.

Maximum Allowable HA (hourangle) - A positive number for the maximum HA permitted (e.g. 5.5 = a HA of +/- 5.5 hours from the Meridian). Calculations for declinations other than 0 will automatically be performed using this input.

Allowable Tracking Near the Meridian - Enter a number for the minutes tracking near the Meridian when the scope is West on the pier and pointing East (e.g. -.5 will stop tracking when the mount reaches 30 minutes of the Meridian).

When a crash is avoided by these features you must take the following actions to reinitialize your mount and the Temma driver:

1. Disconnect the driver
2. Enter the Temma Driver's setup and check the 'Disable Mount Safety Limits' checkbox
3. Reconnect to the Mount
4. Move the mount to a safe position
5. Enter the Temma Driver's setup and uncheck the 'Disable Mount Safety Limits' checkbox



Safety Limits	
Minimum Allowable Altitude	0
Maximum Allowable HA	0
Tracking Near The Meridian Limit	0
☒ Disable Mount Safety Limits	
☐ Warn Before Meridian Flip	

6. Resume

Checking the 'Disable Mount Safety Limits' button will deactivate this feature.

If you want to be warned before the mount flips check the 'Warn..' button.

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Initialization

Temma Setup and Initialization Procedure

1. Turn off the Temma telescope mount.
2. Connect the RS232 serial port on the Temma to an open serial port on your computer.
3. Turn the Temma Power switch on. Each time the Temma is powered on, the Temma control system is initialized with the following default parameter:
 - * The optical tube assembly (or telescope) is assumed to be on the West side of the mount.
 - * The telescope is pointing to 0.00 in both right ascension and declination.
 - * Right ascension and declination correction speeds are set to 90 (unless you have entered a different rate in the Temma driver's main window).
 - * The control system tracking is for the Northern Hemispheret.
4. Open your planetarium software.
5. In the 'ASCOM Telescope Chooser Control list, select **Temma by Takahashi**.
6. Click **Settings** and then select the correct COM port and input your info in the other fields (e.g.: longitude/latitude).
7. Establish a link to the Temma.
8. Select one of the orientation modes from the 'Init' windows
9. When the link is established, center the appropriate star as directed and then synchronize the planetarium software (e.g.: for TheSky press its 'Sync' button).

Synchronizing

Procedure

To synchronize with the optical tube assembly on the East side of the mount, Scope pointing West

1. Loosen the clutches on the mount and move the telescope to the East side of the mount.
2. Center a known star that is on the West side of the Meridian in the field of view.

	3. Identify the star in your planetarium software. 4. Sync on the star. 5. An on screen dialog will notify you of the correct side of the meridian to select the sync star.
To synchronize with the optical tube assembly on the West side of the mount, Scope pointing East	1. Loosen the clutches on the mount and move the telescope to the West side of the mount. 2. Center a known star that is on the <i>East</i> side of the Meridian in the field of view. 3. identify the star in your planetarium software. 4. Sync on the star. 5. An on screen dialog will notify you of the correct side of the meridian to select the sync star.
Scope Pointed at Pole, Counter-weight down position	This method is intended for remote users. The CW down method gives a rough initial alignment that must be refined by a subsequent sync for accurate GOTO operations. It is recommended that the subsequent sync be accomplished using ACP (Pinpoint) to plate solve a CCD image after the initial alignment. Obviously, a manual/visual sync can be used if desired.
Scope Pointed at Pole, Counter-weight West	The CW West method gives a rough initial alignment that must be refined by a subsequent sync for accurate GOTO operations. Place the CW on the West side of mount with the CW shaft and OTA parallel to the ground.
Keep Last Sync	The Temma telescope was previously synced and not powered off, or the Temma was 'Parked'
	New feature: 'Sync Z' allows the user to initialize and accurately sync the mount in the Takahashi 'pointed at the Zenith' position. To use this feature first turn off tracking, point the scope at the Zenith (using a level helps position the scope accurately), then click the 'Sync Z' button.
SyncZ	* power on the mount * start the driver and turn off tracking * polar align * move the scope manually to point to Zenith (use a level for best results) * press SyncZ and the mount starts with RA=sidereal time and Dec=Latitude

Stunning User Interface

Mount Configuration

Mount Type: Select the mount type that matches your mount from the drop-down box.

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Offset Scale

Displays the proper offset for the polar scope based on the users site.

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Site Setup

Site Setup

Latitude: Latitude of the selected site. Latitude must be in degrees: minutes: seconds format.

Longitude: Longitude of the selected site. Longitude must be in degrees: minutes: seconds format.

Elevation: Elevation in meters

To get Longitude/Latitude from TheSky: clicking the button will retrieve the longitude and latitude from TheSky (check the 'X' box if you are using TheSky X). The Longitude/Latitude fields will be populated with the data from TheSky's site data .

Site Information Setup

Latitude N 26 11 15

Longitude W 80 15 15

Elevation (m) 3

TheSky ☐

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Telescope

Aperture: Enter the primary telescope's aperture in millimeters.

Focal Length: Enter the primary telescope's focal length in millimeters.

Note: these fields are not specifically used by the driver but they are passed up to clients if the client asks for them.

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Temma Setup Dialog

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Flip Mount

CAUTION! Please use care and common sense when flipping the mount or using the Meridian Delay feature to ensure that your telescope equipment will not strike your pier.

Care is needed when slewing to objects on the 'wrong' side of the mount. Please test your equipment and determine the safe limits of travel and do not exceed them before attempting to do a Meridian flip of use the Meridian delay feature. Training and using the mount safety feature of the driver is recommended.

The 'Flip Mount' button , when pressed, will prompt the user to perform a manual Meridian Flip. This is very useful to place the mount in a position that allows the longest time for astroimaging. For example, a GOTO to an object that is near and has not yet crossed the Meridian will leave the mount positioned with the OTA on the West side of the pier. Pressing the 'Flip Mount' button will move the mount 180 degrees to position the OTA on the East side of the pier allowing the longest possible time for astroimaging of the object.

Procedure for flipping:

1. Click the 'Flip Mount' button the select 'Yes' - the mount will begin slewing and flip
2. Once the slew completes click the Flip Mount' button again then select 'No'
3. Use your planetarium program to sync to the current position.

Flipping can also be accomplished by client software and scripts by calling the ASCOM SideofPier property.

Note: flipping the mount will automatically invoke the [Meridian Delay](#) feature to ensure subsequent small slews are in the right direction and do not cause the mount to re-flip.

Guide Correction Speeds

Guide Rates

Set

Enter the right ascension and declination correction speed (as a percentage of the Sidereal rate, 10 - 99%). This setting is used for the slow speed moves or for guiding.

High Precision Slew

High precision slew

Check the 'Hi-P Slew' checkbox (marked by red arrow below) to activate this feature to reduce the Temma's inherited encoder resolution errors after a GOTO slew. This offers an automated centering that may be faster and better than doing manual corrections to bring objects to the of the FOV.

Special thanks to Hartmut Bornemann for supplying the code snippets and support implementing this new feature.

Meridian Delay

CAUTION! Please use care and common sense when flipping the mount or using the Meridian Delay feature to ensure that your telescope equipment will not strike your pier.



Care is needed when slewing to objects on the 'wrong' side of the mount. Please test your equipment and determine the safe limits of travel and do not exceed them before attempting to do a Meridian flip or use the Meridian delay feature. Training and using the mount safety feature of the driver is recommended.

Temma mounts have the capability to track through the meridian. This means that an imaging sequence can often be started with the telescope well to the east of the mount (and even under the mount, with the weights above the mount). As the mount tracks it will cross over to the 'right' side of the mount, as it crosses the meridian. This allows imaging sequences that can be completed without ever having to stop and flip the mount. With this capability a Temma mount can maximize imaging time.

The Temma ASCOM V2 driver now has 'Meridian Delay' which enables the mount to slew to objects on the 'wrong' side of the mount, as shown in the above photo. There is a great advantage to this, as described above. Of course, not all objects can be reached in this way - there are two primary limitations; the length of the OTA / CCD camera train, and the declination of the desired object. Generally, the shorter the OTA and the further south the object (towards the celestial equator) then the further east the mount can go, and vice versa. I can image objects in the Southeastern sky with a Takahashi FSQ-106N with the counter weights pointing up.

Using the Meridian Delay feature:

1. Verify the Eastern limits of your equipment
2. Determine the Hour-angle of the the Eastern most point you can safely reached at all Declinations
3. Input that absolute value (no negative) in the Meridian Delay box on the Setup Window. Note, slightly under-estimate your limit then add a very small amount to that to allow plate solving corrections to center objects without reaching the limit. If the limit is exceed the mount will flip back to safely slew to the object you selected outside of Meridian delay limit.
4. You should now be able to safely slew to objects within your tested range without flipping.
5. Entering a '0' (zero) in the Meridian Delay box will disable this feature.

Note: When the Meridian Delay feature is active slews to the Western side will be limited near the horizon by the same amount enter for your Meridian Delay.

Parking

Parking the Temma

The Temma's electronics do not have the ability to park the mount via its firmware. However, this driver offers a solution via software.

Park Procedure:

Two park methods are available:

'Slew Before Park' slews to the inputted alt/az position before parking
 'Park at Current Position' parks the current alt/az position of the mount.



Prior to using the park feature the mount must be correctly

	synced as described above.
--	----------------------------



	The 'Set Park' button can be used to select the mount's current position as the parking position whenever the Temma is connected. The position will be saved thus removing the need to manually input the values.
--	---

1. Input the desired park altitude and azimuth
2. Select one of the park methods
3. Click the 'Park Mount' button
4. When the park slew is completed disconnect the driver and immediately power off the mount or the mount can be left parked.

However, if the power is not turned off the mount will remain in the simulated 'tracking off' mode for NON-Temma 2 models.



	Upon reconnection to the mount the unpark sequence will begin.
--	--

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Power Voltage

Displays the mount's voltage.

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Save Current Position

The 'Save Pos.' button is used to set the current RA and Dec, when the mount is moved via the hand paddles clicking the 'Slew Prior' button will return the mount to the previous position.

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Slew to Prior Coordinates

Clicking The 'Slew Prior' button will return the mount to the position last reported before the current slew.

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Syn to Alt/Az

The 'Sync to Alt/Az' and the 'Slew Prior' button is used to re-sync the mount to the Altitude and Azimuth shown in the user input boxes. This feature is useful when the mount appears to be lost after an unpark.

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Sync To Zenith

'Sync Z' allows the user to initialize and accurately sync the mount in the Takahashi 'pointed at the Zenith' position.

To use this feature:

- * power on the mount
- * start the driver and turn off tracking
- * polar align
- * move the scope manually to point to Zenith (use a level for best results)
- * press SyncZ and the mount starts with RA=sidereal time and Dec=Latitude

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Tracking Rates

Tracking Rates

- Sidereal (default)
- Lunar - When used in conjunction with the 'TheSky Format' checkbox the lunar rate will be set to a high precision mode that will control both the RA speed and Dec rate to account for parallax
- Solar
- King - Rate will compensate for atmospheric refraction
- Manual - When selected rates must be inputted into the RA and DEC boxes, then press the 'Send' button
- Custom - This feature requires TheSky v6. When selected users can input the objects name and update interval in to the 'Custom Tracking Rates' frame. Updates to current offsets will be made at the desired intervals.

Clicking on the 'Tracking Rate' option buttons will toggle the rates.

'Tracking On/Off' is used to stop/start the tracking on Temma 2 mounts, or slow the tracking rate to

~60% sidereal rate for **non** Temma 2 mounts. This is useful if you want to pause viewing. The slower rate will avoid the mount running in to itself when left unattended. Checking the 'Rewind RA' box in the Misc frame (when using **non** Temma 2 mounts) will rewind the right ascension back to the point where the tracking was stopped adding one more level of mount protection for unattended pauses in observing or imaging.

Custom/Comet Tracking Rates

RA Adjustment (sidereal sec/day)

- Enter the adjusted tracking rate for the right ascension axis in *sidereal* seconds per day max = +/- 32767. There are 86,164 seconds in one *sidereal* day (1 "earth day" = 1.002738 "sidereal day").

Dec Adjustment (min/day)

- Enter the adjusted tracking rate for the declination axis in minutes per day, max = +/- 1180 (plus = North, - = South).

Example:

1. +140, +50 - Entering **+140** into the **RA** text box will speed up the RA tracking by 140 seconds per day (one revolution normally takes 86164 seconds at sidereal rate), +50 into the **Dec** text box will move the declination axis 50 minutes per day North. Check the 'Manual Input' box 'Send' to apply the new comet settings. The maximum offsets are +/- 32767 for RA, and +/-1180 for Dec

TheSky Format

- **Checking this box allows for the direct input of RA/Dec rates obtained from Thesky's object info box**
- Input the name of the object (as recognized by Thesky) and the data will be inputted and used automatically to calculate custom rates when the 'Use Thesky' box is checked.

Troubleshooting

Troubleshooting:

The telescope crosshair in my	The coordinates reported by the Temma control
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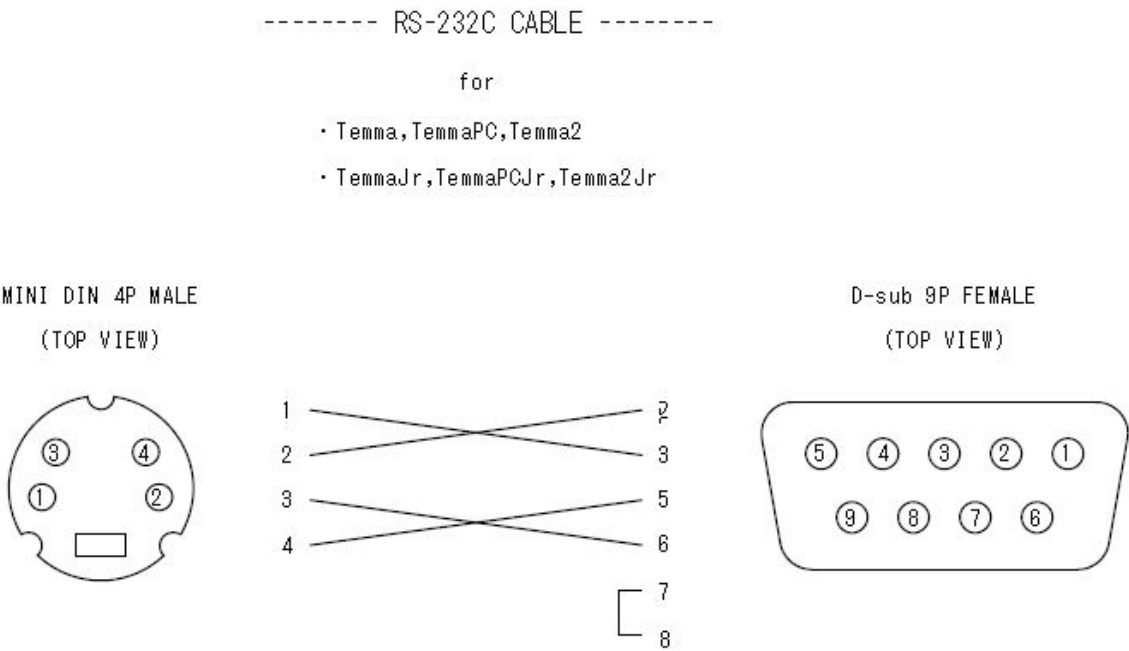
planetarium software jumps	system (when the telescope is not slewing) can vary by one arcsecond. This causes the telescope cross hairs to "jump" between the reported positions. The Temma protocol reports coordinates to the nearest second in right ascension (15 arcseconds at the 0 degrees declination) and to the nearest 6 arcseconds in declination. The Temma's control system accepts declination coordinates to the nearest 12 arcseconds, so, at small fields of view, the cross hairs will not exactly match the coordinate that was used to synchronize the telescope These are limitations of the Temma protocol.
The telescope crosshair in TheSkyX does not move when slewing or syncing	Increase the crosshair update frequency to 1000 ms or more.

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Resources

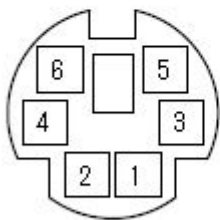
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Cable Drawings



----- AUTO GUIDE CONNECTOR -----

- USD,USD II ,USD3
- Temma,TemmaPC,Temma2
- TemmaJr,TemmaPCJr,Temma2Jr
- PD-7XY,PD-8XY



MINI DIN 6P FEMALE
(TOP VIEW)

No.	
1	NC
2	RA(+)
3	RA(-)
4	DEC(↑)
5	DEC(↓)
6	GND

Driver Updates

Get the most current driver [here](#).